THE QC COVERINGS ON THE CATEGORY OF LOCALLY QUASI-COMPACT HAUSDORFF SPACES FORM A TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument for the three parts, including the quasi-compactness reduction inside part (2) that the author writes out in full.

Editorial note. The label lemma-qc occurs in no other staged Stacks chapter, so no cross-chapter restatement had to be weighed for this card. In particular there is no sibling statement of the same name in the chapter etale-cohomology.tex.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- The category LC of Hausdorff and locally quasi-compact spaces. Same chapter, the definition of the category LC with morphisms being continuous maps. Supplies the ambient category on which the qc coverings are declared.
- qc covering. Same chapter, the definition of a qc covering: a family {f_i : X_i -> X} such that every point of X has a neighbourhood covered by finitely many quasi-compact subsets of the X_i. Supplies the notion verified in all three parts, and the exact shape of the neighbourhoods used in parts (2) and (3).
- Quasi-compactness is stable under fibre products in LC. Same chapter, the permanence of quasi-compactness used when the base change F x_X E_j is called quasi-compact. Supplies the quasi-compactness of F x_X E_j inside X' x_X X_{i_j} in part (3).

Lemma 0.1. *Let X be a Hausdorff and locally quasi-compact space, in other words, an object of LC.*

(1) *If $X' \to X$ is an isomorphism in LC then $\{X' \to X\}$ is a qc covering.*
(2) *If $\{f_i : X_i \to X\}_{i \in I}$ is a qc covering and for each i we have a qc covering $\{g_{ij} : X_{ij} \to X_i\}_{j \in J_i}$, then $\{X_{ij} \to X\}_{i \in I, j \in J_i}$ is a qc covering.*
(3) *If $\{X_i \to X\}_{i \in I}$ is a qc covering and $X' \to X$ is a morphism of LC then $\{X' \times_X X_i \to X'\}_{i \in I}$ is a qc covering.*

Proof. Part (1) holds by the remark above that open coverings are qc coverings.

Proof of (2). Let $x \in X$. Choose $i_1, \ldots, i_n \in I$ and $E_a \subset X_{i_a}$ quasi-compact such that $\bigcup f_{i_a}(E_a)$ is a neighbourhood of x. For every $e \in E_a$ we can find a finite subset $J_e \subset J_{i_a}$ and quasi-compact $F_{e,j} \subset X_{ij}$, $j \in J_e$ such that $\bigcup g_{ij}(F_{e,j})$ is a

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neighbourhood of e. Since E_a is quasi-compact we find a finite collection $e_1, \ldots, e_{m_a}$ such that

$$E_a \subset \bigcup\nolimits_{k=1,\ldots,m_a} \bigcup\nolimits_{j \in J_{e_k}} g_{ij}(F_{e_k,j})$$

Then we find that

$$\bigcup\nolimits_{a=1,\ldots,n} \bigcup\nolimits_{k=1,\ldots,m_a} \bigcup\nolimits_{j \in J_{e_k}} f_i(g_{ij}(F_{e_k,j}))$$

is a neighbourhood of x.

Proof of (3). Let $x' \in X'$ be a point. Let $x \in X$ be its image. Choose $i_1, \ldots, i_n \in I$ and quasi-compact subsets $E_j \subset X_{i_j}$ such that $\bigcup f_{i_j}(E_j)$ is a neighbourhood of x. Choose a quasi-compact neighbourhood $F \subset X'$ of x' which maps into the quasi-compact neighbourhood $\bigcup f_{i_j}(E_j)$ of x. Then $F \times_X E_j \subset X' \times_X X_{i_j}$ is a quasi-compact subset and F is the image of the map $\coprod F \times_X E_j \to F$. Hence the base change is a qc covering and the proof is finished. $\square$